

MODULE - 12

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Subject: Computer Science

Paper: Computer Graphics and Visualization

Module: 3D Object Representations (Bezier, B-Splines)

Module No: CS/CGV/12

Quadrant 1– e-text

Objectives:

- Understand the properties of Bezier curves
- Understand the theory behind B-Splines

Discussion:

Properties of Bezier curves:

From the previous module we have understood how approximating spline curves are drawn. Also Bezier curves were introduced in that module. In this module let's start our discussion with the properties of Bezier curves. Let's recollect the equation for cubic Bezier curve as given

$$B(u) = \frac{1-u}{3} \left[\frac{1+2u}{2} P_0 + \frac{1-2u}{2} P_1 \right] + \frac{u}{3} \left[\frac{1+2u}{2} P_2 + \frac{1-2u}{2} P_3 \right]$$

$$B(u) = \frac{1}{6} (1-u)^3 (2P_0 + P_1) + \frac{1}{6} (1-u)^3 (P_2 + 2P_3) + \frac{1}{2} (1-u)^2 u (P_0 + P_1) + \frac{1}{2} (1-u)^2 u (P_2 + P_3) + \frac{1}{2} u^2 (1-u) (P_1 + P_2) + \frac{1}{2} u^2 (1-u) (P_3 + P_0)$$

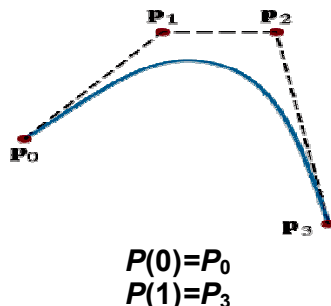
We can write general Bezier curve equation in for n control points, as

$$B(u) = \sum_{i=0}^n \binom{n}{i} (1-u)^{n-i} u^i P_i$$

Property: 1

The curves touch (interpolate) the first and last control points but approximate all other intermediate points.

When we substitute $u=0$ the curve touches the first control point P_0 , and when $u=1$, the curve touches the last control point P_3 .



Property: 2

The curves always lie within their convex hull.

This property means that the curves never lie outside the convex shape of the control polyline. This is because the weights associated with each of the control points is in the interval [0,1] for all values of u in the interval [0,1]. As we know from the previous property that, whenever a curve touches a control point, the weight associated with that point is 1, while all other weights remain insignificant.

Property: 3

The blending functions are non-negative, sum to unity for every u in the range [0,1].

This property means that for any value of u in the interval [0,1] all the blending functions sum to unity.

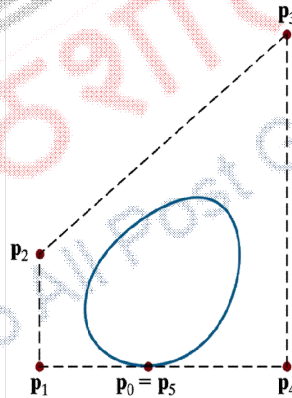
$$\sum_{k=0}^3 BEZ_{k,3}(u) = 1 \quad \forall 0 \leq u \leq 1$$

Property: 4

The slope at the beginning of the curve is along the line joining the first two control points and the slope at the end of the curve is along the line joining the last two end points. Assume there are 6 control points from P_0 to P_5 in the arrangement as shown below, then

$$P'(0) = -5P_0 + 5P_1$$

$$P'(1) = -5P_4 + 5P_5$$



Property: 5

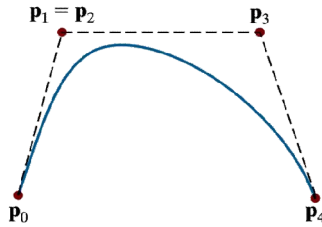
Closed Bezier curves can be generated by specifying the first and last control points at the same position (as shown in the figure above).

As we know from property-1, that the curves touch the first and the last control points, and if we keep both the first and the last control points at the same position the curve becomes closed.

Property: 6

Specifying multiple control points at the same location gives more weight to that position and the curve gets pulled towards them.

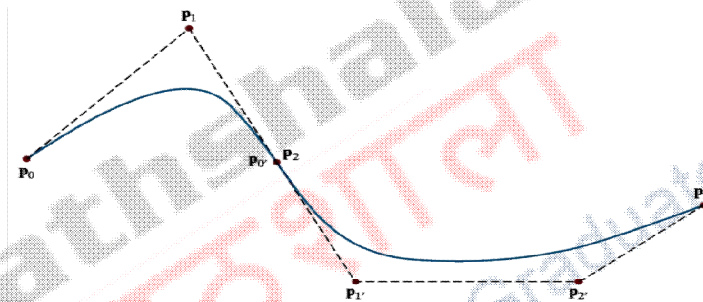
This property means that, when we place more than one control point at the same location the weight at the point tends to be more and the curve gets pulled towards that more than that towards other points. As we know that each control point is associated with one blending function, and with the position being the same the weights get added up and thus the weight is more at that position. This property is useful to design curve shapes of our choice.



Property: 7

Complicated curves can be generated by piecing several Bezier sections of lower degree together.

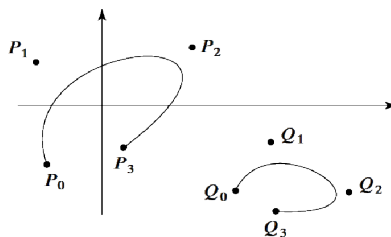
Assume two Bezier curve sections formed by control points, P_0 to P_2 and P_0' to P_3' . Set the initial point of the new curve section as shown below, i.e. $P_0' = P_2$. For achieving C^1 continuity place the second control point of the new section at $P_2 + (P_2 - P_1)$. This property is useful because, Bezier curves are dependent on the number of control points, and in turn on the degree of the polynomials. The more the number of control points, the higher is the degree of the polynomials.



Property: 8

Bezier curves show affine invariance.

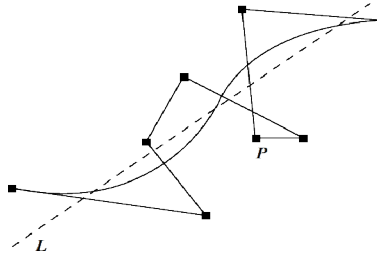
This property states that, Bezier curves are invariant under affine transformations. It can be understood that, if the Bezier curve is to be transformed using affine transformations like Translation, Rotation, Scaling etc., it is as simple as applying these transformations on control points and redrawing the curve at the new position.



Property: 9

Bezier curves do not vary as much as their control polyline does.

This property states that, even though the control polyline varies too much due to random positioning of the control points, the variation in the curve shape is not proportional. This is an interesting property that we tend to ignore, but it signifies the invariance property of the curve even under complex conditions.



Property: 10

If the control points are collinear, Bezier curve reduces to a straight line.

This property states that, if we place the control points along a straight line, Bezier curves cease to be a curve and become a straight line.

Bezier Matrix:

Bezier curve can be represented in matrix form as below.

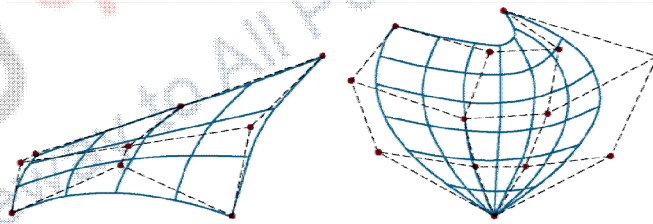
$$P(u) = [u^3 \ u^2 \ u \ 1] \cdot M_{BEZ} \cdot \begin{bmatrix} P_0 \\ P_1 \\ P_2 \\ P_3 \end{bmatrix}$$

Recall the discussion from the previous module (Mod-11) on Hermite Interpolation splines, and if we follow similar solution we get

$$M_{BEZ} = \begin{bmatrix} -1 & 3 & -3 & 1 \\ 3 & -6 & 3 & 0 \\ -3 & 3 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

Bezier Surfaces:

If we extend Bezier curves in two different directions using two parameters say u and v , we get Bezier Surfaces.

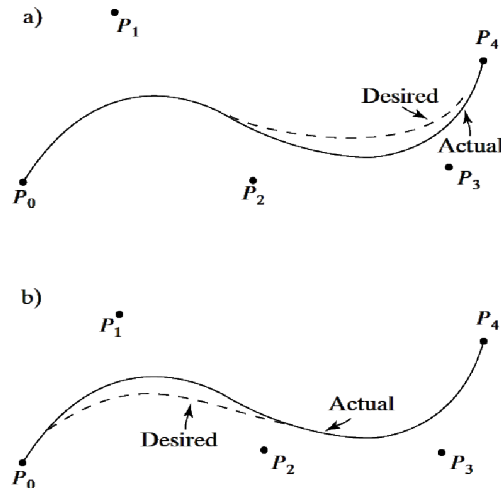


$$P(u, v) = \sum_{j=0}^m \sum_{k=0}^n p_{j,k} BEZ_{j,m}(v) BEZ_{k,n}(u)$$

Drawbacks of Bezier curves:

- They do not offer local control
- The degree of the polynomials is dependent on the number of control points.

Let's understand each of the drawbacks in detail. The first drawback says that Bezier curves do not offer Local Control. It means that if we try to modify the shape of a small section of the curve without disturbing the remaining part of the curve, it would not be possible. What actually happens is that, the rest of the curve also gets affected as shown below in the figure. As can be observed from the figure below, if we try to alter the shape of the curve between P_2 and P_4 , it also affects the curve shape between P_0 and P_2 . Local control is a very important property that determines the freedom of curve design.

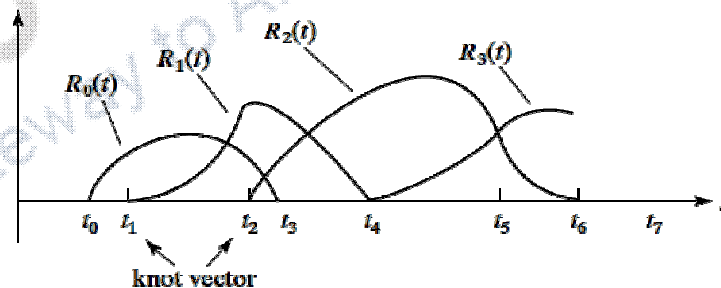


The second drawback says that, the degree of the polynomials is directly related to the number of control points as per the relation

$$\text{Degree} = \text{no. of control points} - 1$$

This dependency should be broken, because, with the parameter u in the interval $[0,1]$, and the powers of u (degree) raised in proportion to the number of control points, it becomes difficult to manage such large floating point values, which is cumbersome often.

Now we need to look for blending functions that satisfy our requirements as mentioned above. These blending functions are key to curve design, as they contribute to the curve shape and also offer better control. Let's expand the range of the parameter from $[0,1]$ to $[-\infty, +\infty]$, the parameter is conventionally referred to as t instead of u . The value of the parameter t_0 where two blending functions meet is called a *knot*. There can be a set of knot values, called *knot vector*, on which we can define a set of blending functions as shown in the figure below.



Thus the requirements for better blending functions are as follows:

- Blending functions must *sum to unity* for every t
- They must be *lower degree* polynomials
- They must offer better *local control*
- They must be *active* over a small span of the entire range of the parameter t
- The degree of the polynomials should be *independent* of the number of control points.

Given a set of knot values, there exists an infinite collection of blending functions, that satisfy the above properties, and there is only one such family that offers the best local

control and has lower degree polynomials, and this family called the Basis for all Spline curves that can be generated.

B-Spline Curves:

The B-Spline curves are defined as follows: Here B stands for $\hat{B}$ asis $\hat{o}$

$$C(t) = \sum_{k=0}^L P_k N_{k,m}(t) \text{ where } t_{k-1} \leq t \leq t_k$$

t	$\ddot{}$	is the parameter
t_k	-	are the knot values (k from min to max)
P_k	$\ddot{}$	control points
0 to L	$\ddot{}$	$L+1$ control points / $L+1$ blending functions
$N_{k,m}(t)$	$\ddot{}$	K^{th} B-spline blending function of order m
m	$\ddot{}$	order of blending functions (one higher than the degree)
$m-1$	$\ddot{}$	degree of blending functions

Blending functions $N_{k,m}(t)$ are derived using Cox-DeBoor recursion formulae as below

$$N_{k,m}(t) = \frac{t - t_{k-1}}{t_k - t_{k-1}} N_{k,m-1}(t) + \frac{t_{k+1} - t}{t_{k+1} - t_k} N_{k+1,m-1}(t)$$

It is evident from the above recursive equation for a B-spline blending function, that a k^{th} B-spline blending function of order m is a weighted combination of two B-spline blending functions of order $m-1$. Thus the above equations are recursive equations and because the minimum possible order is 1, we need to define the first order B-Spline blending functions, and they are defined as follows

$$N_{k,1}(t) = \begin{cases} 1 & t_{k-1} < t \leq t_k \\ 0 & \text{elsewhere} \end{cases}$$

A *knot vector* is a list of parameter values, or knots, that specify the parameter intervals for the individual curves that make up a B-spline.

$t_0, t_1, t_2, \dots, t_{i-1}, t_i$ are $i+1$ non-decreasing numbers.

We can define a *knot vector* with spacing between knots that is either uniform or non-uniform or may have some multiple knots. A Knot vector, with uniform spacing between knots is called a *uniform Knot vector*. Spline curves defined on a *uniform knot vector* are called *Uniform B-Splines*. A knot vector, that is non-uniform, allows placing of knots such that two closely spaced knots might be combined to form one knot by nullifying the spacing between them. It results in a knot called *multiple knot*. The *multiplicity* of a knot determines how many knots are combined into one. A knot vector that has some knots as multiple knots is called a *standard knot vector*. B-splines curves drawn on standard knot vector are called *Open B-Splines*.

Uniform Periodic B-Splines:

Consider a knot vector that is uniform, T ($t_0 = 0, t_1 = 1, t_2 = 2, \dots$)

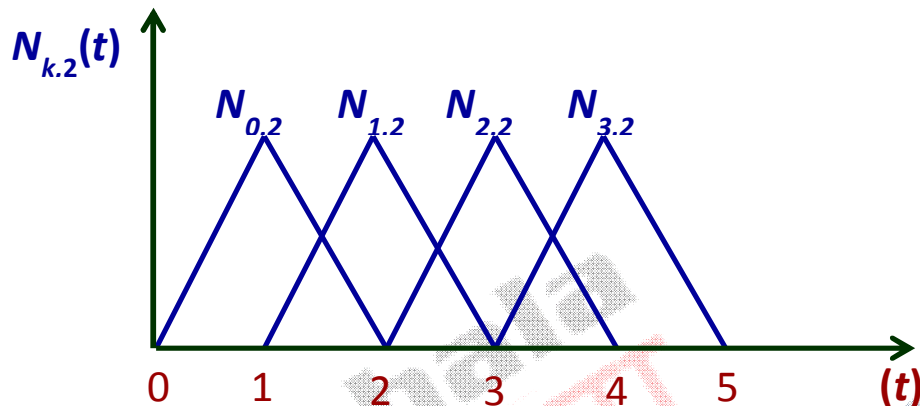
Linear B-Spline blending functions:

Let, $m=2, k=0,1,2,\dots$

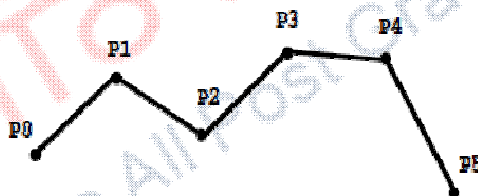
Let's derive the blending functions for this case using Cox-DeBoor recurrence relations as given above.

[illegible]

If we derive equations for other B-spline blending functions of order 2, they simply are shifted versions of the previous functions and have the same shape. It can be observed that $N_{1,2}$ is a shifted version of $N_{0,2}$ by 1 unit to the right on the t axis, similarly $N_{2,2}$ is shifted by 2 units to the right, and so on. We can generalize this by saying that $N_{k,2}$ is shifted by k units to the right. It is also that each function spans 2 units. $N_{0,2}$ has its span between $[0,2]$, $N_{1,2}$ has its span in the interval $[1,3]$ and so on, as shown in the figure below.



A spline curve drawn using blending function of order 2 will have a shape as shown below. In fact, it's not a curve, because the degree of the polynomials is 1. Thus the curves are simply straight lines.



A linear B-spline curve

Quadratic B-Spline blending functions:

We need to set the following

$$m=3, k=0,1,2, \dots$$

$$N_{k,3}(t) = \frac{(t - t_{k-2})(t - t_{k+1})}{(t_k - t_{k-2})(t_{k+1} - t_k)} N_{k-2,2}(t) + \frac{(t - t_{k-1})(t - t_{k+2})}{(t_k - t_{k-1})(t_{k+2} - t_{k+1})} N_{k+1,2}(t) + \frac{(t - t_{k-1})(t - t_{k+1})}{(t_k - t_{k-1})(t_{k+1} - t_k)} N_{k,2}(t)$$

The above equation can be expanded till the order in the terms on the right hand side, reduces to 1. Now substitute in $N_{0,3}(t)$, equations derived earlier for $N_{0,2}(t)$ and $N_{1,2}(t)$

$$N_{0,3}(t) = \frac{(t - t_{-1})(t - t_2)}{(t_1 - t_{-1})(t_2 - t_1)} N_{-1,2}(t) + \frac{(t - t_0)(t - t_3)}{(t_1 - t_0)(t_3 - t_2)} N_{2,2}(t) + \frac{(t - t_0)(t - t_2)}{(t_1 - t_0)(t_2 - t_1)} N_{1,2}(t)$$

$$N_{0,3}(t) = \frac{(t - 0)(t - 3)}{(1 - 0)(3 - 1)} N_{-1,2}(t) + \frac{(t - 1)(t - 4)}{(1 - 1)(4 - 2)} N_{2,2}(t) + \frac{(t - 1)(t - 3)}{(1 - 1)(3 - 1)} N_{1,2}(t)$$

$$\begin{aligned}
 N_{0,3}(t) &= \frac{t^3}{6} + \left[\frac{t^2 - 2t}{2} - \frac{t}{2} + \frac{t - 2}{2} - \frac{t^2 - 2t}{2} \right] \frac{t}{2} + \frac{t^2 - 2t}{2} \frac{t}{2} \\
 N_{0,3}(t) &= \frac{t^3}{6} + \left[\frac{t^2 - 2t}{2} + \frac{t^2 - 2t}{2} + \frac{t - 2}{2} \right] \frac{t}{2} + \frac{t^2 - 2t}{2} \frac{t}{2} \\
 N_{0,3}(t) &= \frac{t^3}{6} + \left[\frac{t^2 - 2t}{2} - \frac{t}{2} \right] \frac{t}{2} + \frac{t^2 - 2t}{2} \frac{t}{2} \\
 N_{0,3}(t) &= \frac{t^3}{6} + \left[\frac{t^2 - 2t}{2} - \frac{t}{2} + \frac{t}{2} \right] \frac{t}{2} + \frac{t^2 - 2t}{2} \frac{t}{2} \\
 N_{0,3}(t) &= \frac{t^3}{6} + \left[\frac{t}{2} - \frac{t^2}{2} - \frac{t}{2} \right] \frac{t}{2} + \frac{t^2 - 2t}{2} \frac{t}{2}
 \end{aligned}$$

Case (i) $0 \leq t \leq 1$:

$$N_{0,3}(t) = \frac{t^3}{6}$$

Case (ii) $1 \leq t \leq 2$:

$$N_{0,3}(t) = \frac{t}{2} - \frac{t^2}{2} - \frac{t}{2}$$

Case (iii) $2 \leq t \leq 3$:

$$N_{0,3}(t) = \frac{t^2 - 2t}{2}$$

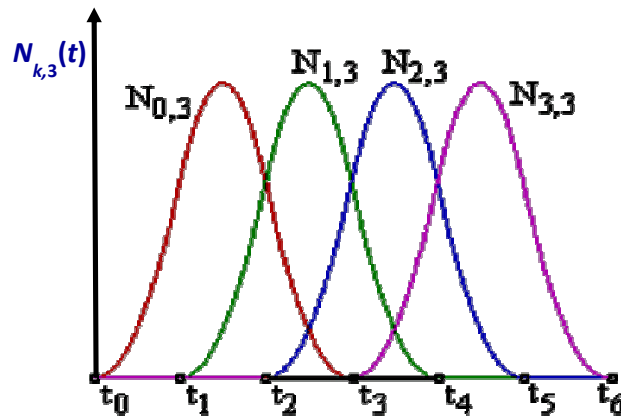
Case (iv) $3 \leq t \leq 4$:

$$N_{0,3}(t) = 0$$

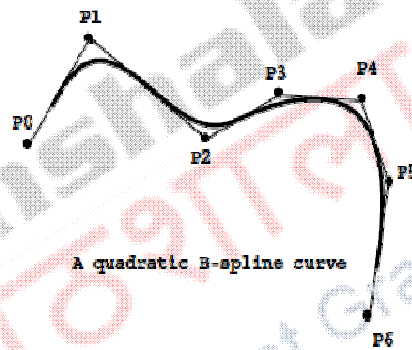
Thus we can define the $N_{0,3}(t)$ blending function as below

$$\begin{aligned}
 N_{0,3}(t) &= \frac{t^3}{6} & 0 \leq t \leq 1 \\
 N_{0,3}(t) &= \frac{t}{2} - \frac{t^2}{2} - \frac{t}{2} & 1 \leq t \leq 2 \\
 N_{0,3}(t) &= \frac{t^2 - 2t}{2} & 2 \leq t \leq 3 \\
 N_{0,3}(t) &= 0 & 3 \leq t \leq 4
 \end{aligned}$$

The shape of blending functions $N_{k,3}(t)$ for k from 0 to n , where n can be any number, is shown as below. It is evident that the shapes are shifted version of the first function $N_{0,3}(t)$. The function $N_{0,3}(t)$ is shifted by 1 unit and has span between [1,4]. The function $N_{2,3}(t)$ is shifted by 2 units to the right and has span between [2,5] and so on. We can conclude in general that a quadratic B-spline blending function $N_{k,3}(t)$ starts at t_k and ends at t_{k+3} . It is easy to derive equations for the other blending functions $N_{k,3}(t)$ by simply substituting for t , in the equation of $N_{0,3}(t)$, $t-k$. (For ex: To derive equations for the blending function $N_{1,3}(t)$, substitute in the equations of $N_{0,3}(t)$, $(t-1)$. To derive equations for the blending function $N_{2,3}(t)$, substitute in the equations of $N_{0,3}(t)$, $(t-2)$ and so on.)



Spline curve drawn using the above quadratic blending functions look like the one as shown below.



Summary:

- We have learnt the important properties of Bezier curves
- Understood the drawbacks of Bezier curves
- Learnt the theory behind drawing Linear, Quadratic, B-Spline curves.
